

Responsible AI Statement

Crop Yield Estimation in Kenya Using Agricultural and Geospatial Data

1. Who does this system affect?

This crop-yield estimation system affects three main groups:

- Agricultural planners, county agricultural officers, researchers, and other users of the model output may use the estimates to understand crop productivity and support agricultural planning.
- Farmers and agricultural communities in the counties represented in the data are the people whose agricultural conditions are being represented by the model, even though individual farmers may never interact with the system directly.
- Smallholder farmers and communities whose local conditions are poorly represented in the available historical data may be affected without having had a direct voice in the design of the model.

The project uses National Agricultural Production Data 2024 covering 2019-2023 and crop-location information for eight food crops, so the model reflects the coverage and limitations of those data sources.

2. What is the worst plausible outcome?

The worst plausible outcome is that an agricultural planner interprets an inaccurate yield estimate as a reliable representation of actual production and uses it to make planning decisions. This risk is particularly important for crops where the model performed poorly. For example, **Pigeon Peas had an R^2 of -0.583**, indicating that the model did not explain the variation in observed yield for this crop well. Relying on such an estimate without considering its uncertainty could therefore lead to inappropriate crop planning, resource allocation, or assessment of agricultural performance.

3. Who bears the cost of errors?

The cost of model errors ultimately falls on **farmers, agricultural communities, and decision-makers who rely on the estimates**, rather than on the model itself. An inaccurate estimate may contribute to inappropriate planning decisions or misrepresentation of the productivity of a particular crop or county. Therefore, the model should not be treated as an independent decision-maker. Its predictions should be interpreted alongside observed agricultural data and reviewed by agricultural professionals, particularly when decisions could significantly affect farming communities.

4. What redress mechanism exists?

At present, this project does not make automated decisions about individual farmers and therefore does not have a formal individual appeal mechanism. If the system were deployed for

high-impact agricultural decisions, users should be able to **question an estimate, request human review, and provide updated or locally verified agricultural information**. Corrections should be documented and considered when the model is evaluated or retrained. Data protection requirements under Kenya's Data Protection Act should also be considered where personal or identifiable information is introduced into a future version of the system.

5. How the model will be monitored

The model should be monitored **at least annually**, when new agricultural production data become available, with additional monitoring if the system is used operationally. Performance should be evaluated using **MAE, RMSE, R², and MAPE**, both overall and separately for each crop and county. A substantial increase in prediction error or a sustained decline in crop-level R² should trigger investigation and potential retraining. A model should not be relied upon for a crop where evaluation shows consistently poor performance until additional data or improved features are available.

Fairness Criterion

The project considers **Individual Fairness**: agricultural observations with similar relevant conditions should receive similar yield estimates, regardless of county identity. This is appropriate because county labels should not by themselves cause substantially different estimates when the underlying environmental and agricultural conditions are similar. However, Individual Fairness was considered conceptually rather than formally quantified in this project, so further work is required to test whether the model consistently satisfies this criterion.

Disaggregated Findings

The Random Forest model produced an overall **MAE of 0.859 t/ha, RMSE of 1.623 t/ha, and R² of 0.863**. However, performance varied substantially across the eight crops:

Crop	N	RMSE	MAE	R ²	MAPE (%)
Beans	40	0.322	0.226	0.017	47.2
Cowpeas	40	0.275	0.219	0.106	38.8
Irish Potatoes	40	2.269	1.912	0.327	29.8
Maize	40	0.463	0.363	0.755	27.7
Pigeon Peas	35	0.359	0.272	-0.583	56.4
Sorghum	40	0.518	0.348	0.091	50.3
Sweet Potatoes	40	3.705	2.820	0.251	48.9
Wheat	28	0.668	0.543	0.564	31.8

The strongest crop-level performance was observed for **Maize and Wheat**, while Pigeon Peas showed particularly poor generalization.

Datasheet Summary

The dataset represents agricultural production for selected high-producing counties and eight food crops using national agricultural production data from **2019–2023**, supplemented by crop-location information. It does not directly represent every individual farmer, farm, agricultural practice, or county equally, and therefore model results should not automatically be generalized to agricultural conditions outside the represented data.